

LSTAT2130 : Introduction to Bayesian Statistics

Group K

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May 30, 2025

1 Question 1

a) – Likelihood of the Model

We consider the following hierarchical model for the number of hospital visits:

- $y_i \mid \mu_i \sim \text{Poisson}(\mu_i)$
- $\mu_i = v_{g(i)} \cdot \exp(\beta_0 + \beta_1 \cdot \text{age}_i + \beta_2 \cdot \text{chronic}_i)$

where:

- $v_{g(i)}$ is the hospital-specific random effect for the hospital $g(i)$ to which patient i belongs,
- $\beta_0, \beta_1, \beta_2$ are fixed regression coefficients.

The likelihood function is:

$$L(y \mid \beta, v) = \prod_{i=1}^n \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!} \propto \prod_{i=1}^n \mu_i^{y_i} e^{-\mu_i}$$

Substituting $\mu_i = v_{g(i)} \cdot \exp(\beta_0 + \beta_1 \cdot \text{age}_i + \beta_2 \cdot \text{chronic}_i)$ gives:

$$\prod_{i=1}^n [v_{g(i)} \cdot \exp(\beta_0 + \beta_1 \cdot \text{age}_i + \beta_2 \cdot \text{chronic}_i)]^{y_i} \cdot \exp(-v_{g(i)} \cdot \exp(\beta_0 + \beta_1 \cdot \text{age}_i + \beta_2 \cdot \text{chronic}_i))$$

Define:

$$\eta_i := \beta_0 + \beta_1 \cdot \text{age}_i + \beta_2 \cdot \text{chronic}_i$$

Then we can write:

$$L(y \mid \beta, v) \propto v_{g(i)}^{\sum_{i=1}^n y_i} \cdot \exp\left(\sum_{i=1}^n (y_i \eta_i - v_{g(i)} \exp(\eta_i))\right)$$

b) – Joint posterior distribution

We are interested in the joint posterior distribution:

$$p(\beta, v_g, \alpha \mid y) = L(y \mid \beta, v) \cdot p(v_g) \cdot p(\alpha) \cdot p(\beta)$$

- The prior on the random effects: $v_g \sim \text{Gamma}(\alpha, \alpha)$ implies:

$$p(v_g \mid \alpha) = \prod_{g=1}^G \frac{\alpha^\alpha}{\Gamma(\alpha)} v_g^{\alpha-1} e^{-\alpha v_g} \propto \prod_{g=1}^G v_g^{\alpha-1} e^{-\alpha v_g}$$

- The prior on the hyperparameter: $\alpha \sim \text{Gamma}(a_0, b_0)$: $p(\alpha) \propto \alpha^{a_0-1} e^{-b_0 \alpha}$
- The priors on regression coefficients: $\beta_j \sim \mathcal{N}(0, \sigma^2 = 100)$ independently, which means:

$$p(\beta) \propto \exp\left(-\frac{1}{2 \cdot 100} (\beta_0^2 + \beta_1^2 + \beta_2^2)\right)$$

The full joint posterior is therefore:

$$p(\beta, v_g, \alpha \mid y) \propto v_{g(i)}^{\sum_{i=1}^n y_i} \cdot \exp\left(\sum_{i=1}^n (y_i \eta_i - v_{g(i)} \exp(\eta_i))\right) \cdot \left[\prod_{g=1}^G v_g^{\alpha-1} e^{-\alpha v_g}\right] \cdot \alpha^{a_0-1} e^{-b_0 \alpha} \cdot \exp\left(-\frac{1}{200} \sum_{k=0}^2 \beta_K^2\right)$$

$$\propto \exp\left(\sum_{i=1}^n (y_i \eta_i) - \frac{1}{200} \sum_{k=0}^2 \beta_K^2 - b_0 \alpha\right) \prod_{g=1}^G \left[v_g^{\sum_{i=1}^n y_i + \alpha - 1} \exp(-\sum_{i=1}^n v_{g(i)} \exp(\eta_i) - v_{g(i)} \alpha)\right] \alpha^{a_0-1}$$

2 Question 2

a) – Full conditionals for the regression coefficients $\beta_0, \beta_1, \beta_2$

We derive the full conditionals for the regression coefficients.

$$p(\beta_0 | v_{g(i)}, \alpha, y) \propto \exp \left\{ \sum_{i=1}^n (y_i \beta_0 - v_{g(i)} e^{\beta_0 + \beta_1 \text{age}_i + \beta_2 \text{chronic}_i}) - \frac{\beta_0^2}{200} \right\}$$

$$p(\beta_1 | v_{g(i)}, \alpha, y) \propto \exp \left\{ \sum_{i=1}^n (y_i \beta_1 \text{age}_i - v_{g(i)} e^{\beta_0 + \beta_1 \text{age}_i + \beta_2 \text{chronic}_i}) - \frac{\beta_1^2}{200} \right\}$$

$$p(\beta_2 | v_{g(i)}, \alpha, y) \propto \exp \left\{ \sum_{i=1}^n (y_i \beta_2 \text{chronic}_i - v_{g(i)} e^{\beta_0 + \beta_1 \text{age}_i + \beta_2 \text{chronic}_i}) - \frac{\beta_2^2}{200} \right\}$$

b) – Full conditionals for the hospital-level random effects v_g

The full conditional for a given v_g depends only on the patients assigned to hospital g :

$$p(v_g | \beta, \alpha, y) \propto v_g^{\sum_{i:g(i)=g} y_i + \alpha - 1} \cdot \exp \left(-v_g \left[\alpha + \sum_{i:g(i)=g} e^{\beta_0 + \beta_1 \text{age}_i + \beta_2 \text{chronic}_i} \right] \right)$$

This is a Gamma distribution:

$$v_g | \beta, \alpha, y \sim \text{Gamma} \left(\sum_{i:g(i)=g} y_i + \alpha, \quad \alpha + \sum_{i:g(i)=g} e^{\beta_0 + \beta_1 \text{age}_i + \beta_2 \text{chronic}_i} \right)$$

c) – Hyperparameter α

$$P(\alpha | \beta, v_g, y) = \left(\prod_{g=1}^G v_g^\alpha e^{-\alpha v_g} \right) \alpha^{a_0 - 1} e^{-b_0 \alpha}$$

3 Question 3 (The full implementation of the MCMC algorithm is available in Appendix 6)

The goal of this section is to sample from the joint posterior distribution of the model parameters $\beta = (\beta_0, \beta_1, \beta_2)$, the hospital-specific random effects $v_1, \dots, v_G$, and the hyperparameter α , using a manually implemented Markov Chain Monte Carlo (MCMC) algorithm in R.

The algorithm combines:

- **Gibbs sampling** for the random effects v_g (conjugate update)
- **Univariate random-walk Metropolis-Hastings** for each component of β
- **Metropolis-Hastings on the log scale** for α

1. Gibbs update for v_g

The full conditional distribution is:

$$v_g | \text{rest} \sim \text{Gamma} \left(\sum_{i \in g} y_i + \alpha, \quad \sum_{i \in g} \exp(\eta_i) + \alpha \right)$$

where $\eta_i = \beta_0 + \beta_1 \cdot \text{age}_i + \beta_2 \cdot \text{chronic}_i$.

2. Metropolis-Hastings for β_j

For each β_j :

1. Propose a new β_j^* around its current value:

$$\beta_j^* \sim N(\beta_j^{(t-1)}, \sigma^2)$$

(This is a univariate random-walk proposal)

2. Calculate the linear predictor with the new value:

$$\eta_i^* = \beta_0^* + \beta_1 \text{age}_i + \beta_2 \text{chronic}_i$$

3. Compute the corresponding likelihood:

$$\mu_i^* = v_{g(i)} \cdot \exp(\eta_i^*)$$

4. Calculate the acceptance ratio:

$$r = \frac{p(y \mid \beta^*)p(\beta^*)}{p(y \mid \beta)p(\beta)}$$

5. Working in log-space:

$$\log r = [\log \text{likelihood}(\beta^*) - \log \text{likelihood}(\beta)] + [\log \text{prior}(\beta^*) - \log \text{prior}(\beta)]$$

6. Accept or reject according to:

$$\min(1, \exp(\log r))$$

3. Metropolis-Hastings on log scale for α

1. **Propose a new value on the log scale:**

$$\log(\alpha^*) \sim \mathcal{N}(\log(\alpha^{(t-1)}), \sigma^2) \quad \Rightarrow \quad \alpha^* = \exp(\log(\alpha^*))$$

This ensures that $\alpha > 0$.

2. **Compute the log of the unnormalized posterior:**

$$\log p(\alpha \mid v) \propto G(\alpha \log \alpha - \log \Gamma(\alpha)) + (\alpha - 1) \sum_{g=1}^G \log v_g - \alpha \sum_{g=1}^G v_g + \log p(\alpha)$$

3. **Include the Jacobian of the transformation:**

$$\log \tilde{p}(\alpha) = \log p(\alpha \mid v) + \log(\alpha)$$

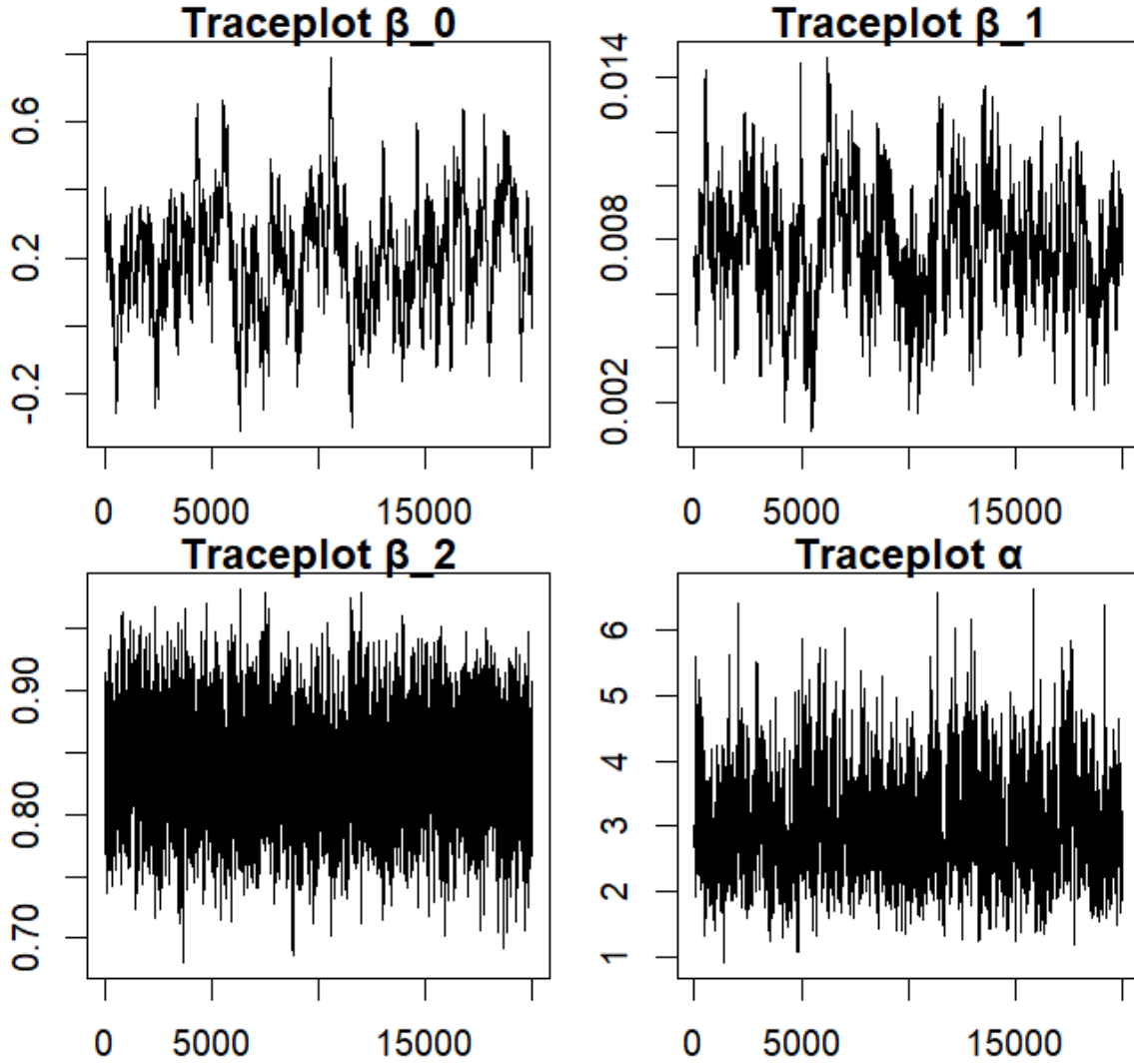
4. **Compute the log acceptance ratio:**

$$\log r = \log \tilde{p}(\alpha^*) - \log \tilde{p}(\alpha^{(t-1)})$$

5. **Accept or reject according to:**

Accept with probability $\min(1, \exp(\log r))$

Otherwise keep the previous value: $\alpha^{(t)} = \alpha^{(t-1)}$.



4 Question 4 (The full implementation of the question 4 is available in [Appendix 6](#))

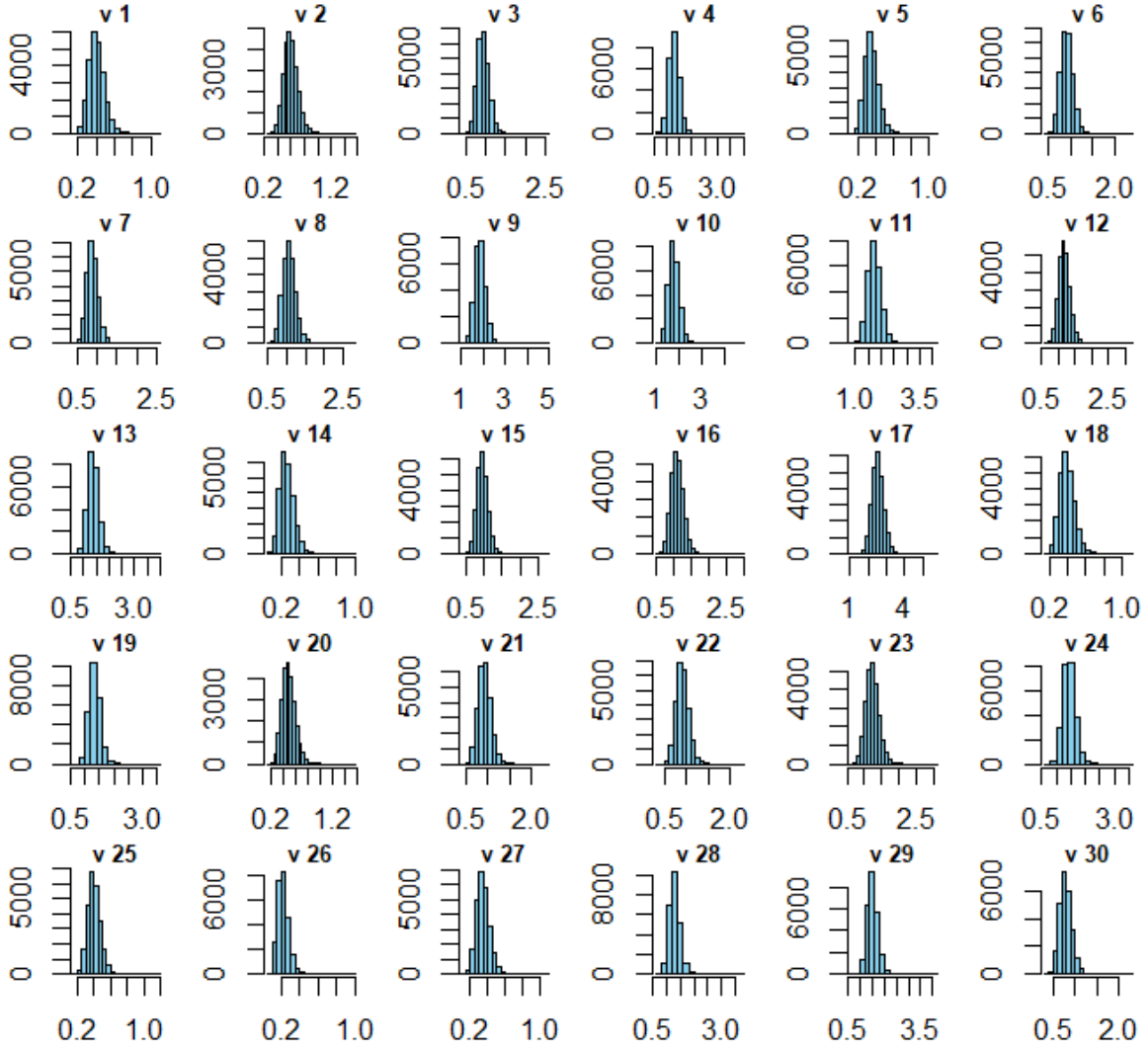
a) –numerical and graphical summaries

This section presents key numerical summaries (posterior means, standard deviations, and credible intervals) for the fixed effects ($\beta_0, \beta_1, \beta_2$) and hyperparameter α quantifying their uncertainty and clinical significance.

Parameter	Mean	Median	sd	2.5%	97.5%	HDI 2.5%	HDI 97.5%
β_0	0.2184	0.2251	0.1523	-0.0802	0.5113	-0.0893	0.4976
β_1	0.0074	0.0073	0.0021	0.0035	0.0117	0.0034	0.0116
β_2	0.8344	0.8339	0.0432	0.7521	0.9210	0.7516	0.9199
α	2.9120	2.8266	0.7603	1.6696	4.6473	1.5204	4.4277

Table 1: Posterior statistics of the parameters

For the hospital-specific random effects ($\nu_1, \dots, \nu_{30}$), we provide graphical representations to visualize between-hospital variability:



b) – Posterior Predictive Performance

To assess the quality of the fitted model we evaluate its posterior predictive performance this is done by comparing the observed data distribution with replicated datasets generated from the posterior predictive distribution:

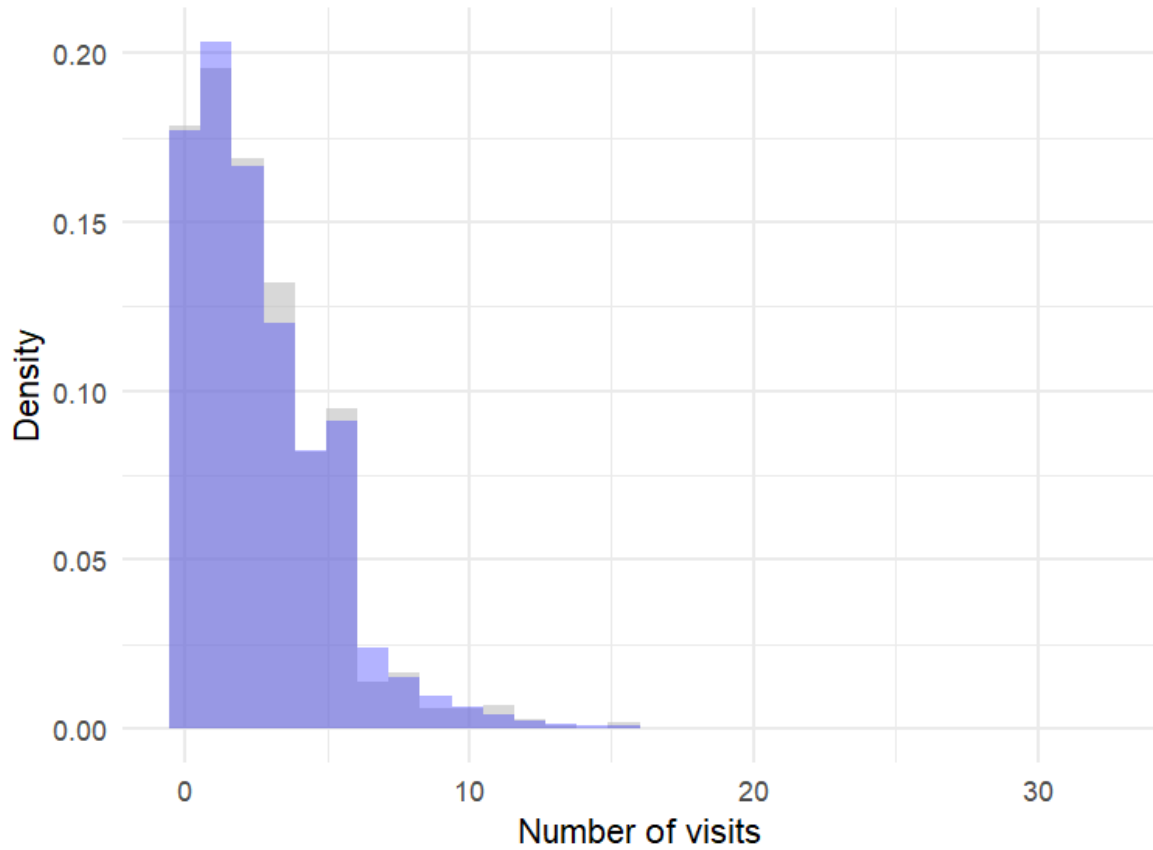
- We sample values of (β, v) from the posterior draws.
- For each sample, we compute the corresponding means μ_i .
- We generate new responses $y_i^{\text{rep}} \sim \text{Poisson}(\mu_i)$.

We then compare the empirical distribution of the observed responses y with the distribution of the replicated responses y^{rep} , both numerically and graphically.

Graphical comparisons

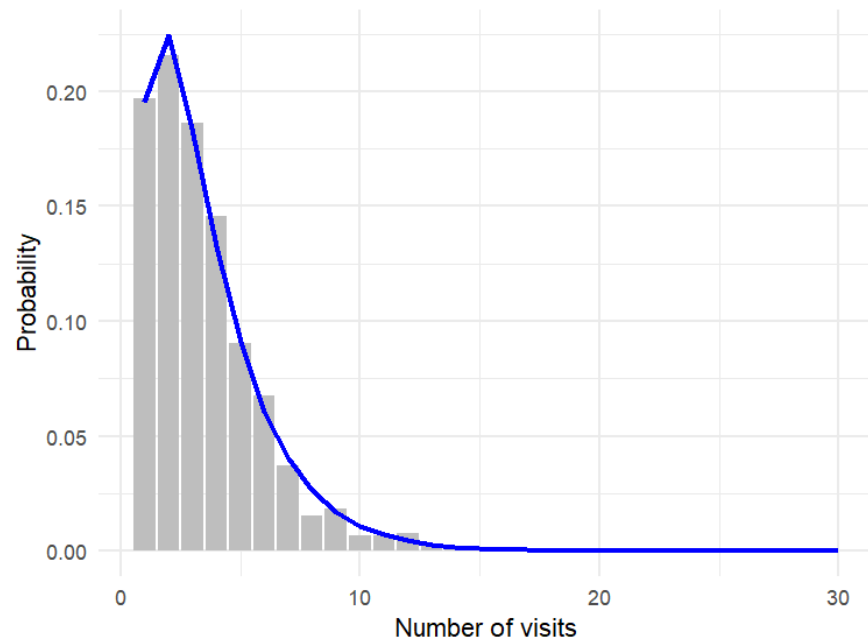
- The histogram comparison shows strong alignment between the observed and predictive distributions.

Comparison of distributions



- The bar plot of empirical vs simulated probabilities confirms the model captures the frequency structure well.

Comparison of observed vs. simulated probabilities



Numerical summaries

Table 2: Posterior Predictive Performance Metrics

Metric	Observed	Posterior Predictive
Mean	2.495	2.531
Variance	5.727	5.994
Coverage		0.992
Bias		0.036

The results demonstrate that the posterior predictive distribution closely aligns with the observed data. The posterior predictive mean (2.531) is very close to the empirical mean (2.495), and the predictive variance (5.994) also accurately reflects the observed variance (5.727). Moreover, the 95% credible interval coverage reaches 99.2% indicating that the model's uncertainty estimates are well-calibrated. The average bias between the simulated and observed responses is minimal (0.036).

5 Question 5 (The full implementation of the question 5 is available in Appendix 6)

a) See the related code in the appendix

b) Comparaison between the two models

In the table below, we observe that our custom MCMC implementation and the JAGS version are highly similar which suggests that choices made in point 3 were appropriate. Indeed, the small deviations between the models (< 0.03 for the means, < 0.02 for the SD) can be explained by the Monte Carlo variability.

Parameter	Mean	Median	sd	2.5%	97.5%	HDI 2.5%	HDI 97.5%
β_0	0.2184	0.2251	0.1523	-0.0802	0.5113	-0.0893	0.4976
β_0^*	0.2407	0.2438	0.1672	-0.1031	0.5603	-0.0833	0.5775
β_1	0.0074	0.0073	0.0021	0.0035	0.0117	0.0034	0.0116
β_1^*	0.0073	0.0073	0.0022	0.0031	0.0116	0.0031	0.0116
β_2	0.8344	0.8339	0.0432	0.7521	0.9210	0.7516	0.9199
β_2^*	0.8348	0.8350	0.0433	0.7497	0.9194	0.7506	0.9202
α	2.9120	2.8266	0.7603	1.6696	4.6473	1.5204	4.4277
α^*	2.8708	2.7897	0.7524	1.6219	4.5450	1.5065	4.3666

Table 3: Posterior statistics of the parameters for the hand-written model and the jags model (*)

6 Question 6 (The full implementation of the question 6 is available in Appendix 6)

In this section, we observe that changing the prior on v_g from gamma to log-normal strongly affects β_0 a lot while β_1 , β_2 stay robust. This means that the baseline hospital visit rate is really sensitive to how the variability between the hospitals is modeled. However, age and chronic illness are robust regarding the prior specifications. In addition, the hyperparameter τ is similar in scale to α suggesting that both models capture almost the same inter-variability.

Parameter	Mean	Median	sd	2.5%	97.5%	HDI 2.5%	HDI 97.5%
β_0	0.0392	0.0368	0.1649	-0.2758	0.3710	-0.2762	0.3703
β_1	0.0073	0.0073	0.0021	0.0033	0.0114	0.0033	0.0115
β_2	0.8353	0.8348	0.0424	0.7531	0.9188	0.7550	0.9205
τ	2.4226	2.3509	0.6844	1.2928	3.9639	1.1878	3.7879

Table 4: Posterior statistics of the parameters

Appendix

This appendix contains the full R code

Question 3: Custom MCMC Sampler

```
set.seed(123)
y <- HospitalVisits$y
age <- HospitalVisits$age
chronic <- HospitalVisits$chronic
hospital <- HospitalVisits$hospital

G <- length(unique(hospital))
n <- length(y)
a0 <- 0.01
b0 <- 0.01
sigma2_beta <- 100
#initialisation
n_iter <- 25000
burn_in <- 5000
beta <- matrix(0, nrow = n_iter, ncol = 3)
colnames(beta) <- c("beta0", "beta1", "beta2")
v <- matrix(1, nrow = n_iter, ncol = G)
alpha <- numeric(n_iter)
alpha[1] <- 1
prop_sd_beta <- c(0.05, 0.001, 0.05)
prop_sd_log_alpha <- 0.1

#MCMC univariate
for (t in 2:n_iter){
  beta_curr <- beta[t-1, ]
  v_curr <- v[t-1, ]
  alpha_curr <- alpha[t-1]
  eta <- beta_curr[1] + beta_curr[2]*age + beta_curr[3]*chronic

  # ---- Gibbs for v_g ----
  for(g in 1:G){
    idx <- which(hospital == g)
    s <- sum(y[idx]) + alpha_curr
    r <- sum(exp(eta[idx])) + alpha_curr
    v[t, g] <- rgamma(1, shape = s, rate = r)
  }

  v_g <- v[t, ]
  mu_curr <- v_g[hospital] * exp(eta)

  # ---- MH for beta ----
  for(j in 1:3){
    beta_prop <- beta_curr
    beta_prop[j] <- rnorm(1, beta_curr[j], prop_sd_beta[j])
    eta_prop <- beta_prop[1] + beta_prop[2]*age + beta_prop[3]*chronic
    mu_prop <- v_g[hospital] * exp(eta_prop)

    log_ratio <- sum(dpois(y, mu_prop, log = TRUE)) -
      sum(dpois(y, mu_curr, log = TRUE)) +
      dnorm(beta_prop[j], 0, sqrt(sigma2_beta), log = TRUE) -
```

```

dnorm(beta_curr[j], 0, sqrt(sigma2_beta), log = TRUE)

if (log(runif(1)) < log_ratio) {
  beta_curr <- beta_prop
  eta <- eta_prop
  mu_curr <- mu_prop
}

}

beta[t, ] <- beta_curr
# ---- MH for log(alpha) ----
log_alpha_curr <- log(alpha_curr)
log_alpha_prop <- rnorm(1, log_alpha_curr, prop_sd_log_alpha)
alpha_prop <- exp(log_alpha_prop)

log_post <- function(a, v_vals) {
  G * (a * log(a) - lgamma(a)) + (a - 1)*sum(log(v_vals)) - a * sum(v_vals) +
  dgamma(a, a0, b0, log = TRUE)
}

log_post_prop <- log_post(alpha_prop, v_g)
log_post_curr <- log_post(alpha_curr, v_g)
log_post_prop <- log_post_prop + log(alpha_prop)
log_post_curr <- log_post_curr + log(alpha_curr)

# Calcul du ratio d'acceptation
prob <- min(1, exp(log_post_prop - log_post_curr))

# Acceptation / rejet
if (runif(1) <= prob) {
  alpha[t] <- alpha_prop
} else {
  alpha[t] <- alpha_curr
}
}

post_beta <- beta[(burn_in + 1):n_iter, ]
post_alpha <- alpha[(burn_in + 1):n_iter]

par(mfrow = c(2, 2))
plot(post_beta[,1], type = "l", main = "Traceplot _0")
plot(post_beta[,2], type = "l", main = "Traceplot _1")
plot(post_beta[,3], type = "l", main = "Traceplot _2")
plot(post_alpha, type = "l", main = "Traceplot ")

```

Question 4: Posterior Analysis and Predictive Performance

#a

```

library(HDInterval)
data_pros<-data.frame(post_beta,post_alpha)
col <- c("beta0", "beta1", "beta2", "post_alpha")

results <- t(apply(data_pros[, col], 2, function(x) {
  c(
    mean = mean(x),

```

```

    median = median(x),
    sd = sd(x),
    quantile(x, c(0.025, 0.975)),
    hdi = as.numeric(hdi(x, credMass = 0.95))
  )
}))
print(results)

par(mfrow = c(5, 6), mar = c(2, 2, 1, 1))

for (g in 1:30) {
  hist(v[, g],
    main = paste("v", g),
    xlab = "",
    col = "skyblue",
    breaks = 20,
    cex.main = 0.8)
}

-----
#b

n_sim <- 3000
n <- length(y)
n_post <- nrow(post_beta)
y_rep <- matrix(NA, nrow = n_sim, ncol = n)

set.seed(123)
ids <- sample(1:n_post, size = n_sim)
#simuler y
for (s in 1:n_sim) {
  beta_s <- post_beta[ids[s], ]
  v_s <- v[ids[s], ]

  eta_s <- beta_s[1] + beta_s[2] * age + beta_s[3] * chronic
  mu_s <- v_s[hospital] * exp(eta_s)

  y_rep[s, ] <- rpois(n, mu_s)
}

#comparaison
df_obs <- as.data.frame(table(y) / length(y))
df_rep <- as.data.frame(table(as.vector(y_rep)) / length(as.vector(y_rep)))
names(df_obs) <- names(df_rep) <- c("value", "prob")
ggplot() +
  geom_col(data = df_obs, aes(x = as.numeric(value), y = prob),
    fill = "grey") +
  geom_line(data = df_rep, aes(x = as.numeric(value), y = prob),
    color = "blue", linewidth = 1) +
  labs(title = "Comparison of observed vs. simulated probabilities",
    x = "Number of visits", y = "Probability") +
  theme_minimal()

ggplot() +
  geom_histogram(aes(x = y, y = ..density..),
    bins = 30, fill = "grey", alpha = 0.6, position = "identity") +

```

```

geom_histogram(aes(x = as.vector(y_rep), y = ..density..),
               bins = 30, fill = "blue", alpha = 0.3, position = "identity") +
labs(title = "Comparison of distributions",
     x = "Number of visits", y = "Density") +
theme_minimal()

cat("Empirical mean:", mean(y), "\n")
cat("Predictive mean:", mean(y_rep), "\n")

cat("Empirical variance:", var(y), "\n")
cat("Predictive variance:", var(as.vector(y_rep)), "\n")

lower <- apply(y_rep, 2, quantile, probs = 0.025)
upper <- apply(y_rep, 2, quantile, probs = 0.975)
coverage <- mean(y >= lower & y <= upper)
cat("Couverture à 95% :", round(coverage, 3), "\n")

bias <- mean(colMeans(y_rep) - y)
cat("Biais :", round(bias, 3), "\n")

```

Question 5

```

library(rjags)

data_jags <- list(y = y, age = age, chronic = chronic, hospital = hospital, N = length(y), G = length(hospital))

para_init <- function() {
  list(
    beta0 = rnorm(1),
    beta1 = rnorm(1),
    beta2 = rnorm(1),
    alpha = rgamma(1, 1, 1),
    v = rgamma(length(unique(hospital)), 1, 1)
  )
}

q5_jags <- "model {
  for (i in 1:N) {
    y[i] ~ dpois(mu[i])
    mu[i] <- v[hospital[i]] * exp(beta0 + beta1 * age[i] + beta2 * chronic[i])
  }

  beta0 ~ dnorm(0, 0.01)
  beta1 ~ dnorm(0, 0.01)
  beta2 ~ dnorm(0, 0.01)

  for (g in 1:G) {
    v[g] ~ dgamma(alpha, alpha)
  }

  alpha ~ dgamma(0.01, 0.01)
}"

```

```

model_q5 <- jags.model(textConnection(q5_jags),
                        data = data_jags,
                        inits = para_inits,
                        n.chains = 1,
                        n.adapt = 1000)

#itérations avec burn-in
samples_q5 <- coda.samples(model_q5, variable.names = c("beta0", "beta1", "beta2", "alpha"), n.iter =
-----
# 5b
summary(samples_q5)
results_jags5 <- t(apply(as.matrix(samples_q5), 2, function(x) {
  c(mean = mean(x),
    median = median(x),
    sd = sd(x),
    quantile(x, c(0.025, 0.975)),
    hdi = as.numeric(hdi(x, credMass = 0.95))
  )
}))

#comparaison
comp2model <- round(rbind(results, results_jags5), 4)
rownames(comp2model) <- c("beta0", "beta1", "beta2", "post_alpha", "post_alpha_jags",
                          "beta0_jags", "beta1_jags", "beta2_jags")

print(comp2model)

```

Question 6

```

q6_jags <- "model {
  for (i in 1:N) {
    y[i] ~ dpois(mu[i])
    mu[i] <- exp(u[hospital[i]]) * exp(beta0 + beta1 * age[i] + beta2 * chronic[i])
  }

  beta0 ~ dnorm(0, 0.01)
  beta1 ~ dnorm(0, 0.01)
  beta2 ~ dnorm(0, 0.01)

  for (g in 1:G) {
    u[g] ~ dnorm(0, tau)
  }

  tau ~ dgamma(0.01, 0.01)
}"

para_inits2 <- function() {
  list(
    beta0 = rnorm(1),
    beta1 = rnorm(1),
    beta2 = rnorm(1),

```

```

    tau = rgamma(1, 1, 1),
    u = rnorm(length(unique(hospital)), 0, 1)
  )
}

model_q6 <- jags.model(textConnection(q6_jags),
  data = data_jags,
  inits = para_inits2,
  n.chains = 1,
  n.adapt = 1000)

samples_q6 <- coda.samples(model_q6, variable.names = c("beta0", "beta1", "beta2", "tau"), n.iter = 20000)

results_jags6 <- t(apply(as.matrix(samples_q6), 2, function(x) {
  c(mean = mean(x),
    median = median(x),
    sd = sd(x),
    quantile(x, c(0.025, 0.975)),
    hdi = as.numeric(hdi(x, credMass = 0.95))
  )
}))
print(round(results_jags6,4))

```